

P0192

Diagnostic information - DME (DFI) control unit

Fuel high-pressure sensor

Diagnostic conditions

- Time since engine start > 5 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Fuel high-pressure sensor faulty

P0195

Diagnostic information - DME (DFI) control unit

Oil temperature implausibly low

Diagnostic conditions

- Engine running
- None of the following faults stored: Engine coolant temperature or oil temperature sensor

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit between lines
- ♦ Oil temperature sensor faulty

Fault setting condition(s)

- Engine oil temperature (T030) < 40 °C (104 °F) at model temperature > 90 °C (194 °F)
- Engine oil temperature (T030) > 120 °C (248 °F) at model temperature < 70 °C (158 °F)

P0197

Diagnostic information - DME (DFI) control unit

Oil temperature sensor - electrical fault

Diagnostic conditions

- Time since engine start > 5 minute(s)
- Coolant temperature -15 ... 115 °C (5 ... 239 °F)
- Oil temperature -39 ... 149 °C (-38.2 ... 300 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Oil temperature sensor faulty

Fault setting condition(s)

♦ P0197:

- Oil temperature (measured value) < -39 °C (-38 °F), with:
 - Engine running > 300 second(s)

or

- Ignition on or engine running
- and
- Coolant temperature > -15 °C (5 °F)

♦ P0198:

- Oil temperature (measured value) > 149 °C (300 °F), with:
 - Ignition on or engine running
- and
- Engine coolant temperature < 115 °C (239 °F)

P0198

Diagnostic information - DME (DFI) control unit

Oil temperature sensor - electrical fault

Diagnostic conditions

- Time since engine start > 5 minute(s)
- Coolant temperature -15 ... 115 °C (5 ... 239 °F)
- Oil temperature -39 ... 149 °C (-38.2 ... 300 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Oil temperature sensor faulty

Fault setting condition(s)

♦ P0197:

- Oil temperature (measured value) < -39 °C (-38 °F), with:
 - Engine running > 300 second(s)

or

- Ignition on or engine running
- and
- Coolant temperature > -15 °C (5 °F)

♦ P0198:

- Oil temperature (measured value) > 149 °C (300 °F), with:
 - Ignition on or engine running
- and
- Engine coolant temperature < 115 °C (239 °F)

P0199

Diagnostic information - DME (DFI) control unit

Oil temperature sensor - electrical fault

Diagnostic conditions

- Time since engine start > 5 minute(s)
- Coolant temperature -15 ... 115 °C (5 ... 239 °F)
- Oil temperature -39 ... 149 °C (-38.2 ... 300 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Oil temperature sensor faulty

Fault setting condition(s)

♦ P0197:

- Oil temperature (measured value) < -39 °C (-38 °F), with:
 - Engine running > 300 second(s)

or

- Ignition on or engine running
and
• Coolant temperature > -15 °C (5 °F)

♦ P0198:

- Oil temperature (measured value) > 149 °C (300 °F), with:
 - Ignition on or engine running
and
• Engine coolant temperature < 115 °C (239 °F)

P0201

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 1

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 1 fuel injector faulty

P0202

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 2

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 2 fuel injector faulty

P0203

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 3

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 3 fuel injector faulty

P0204

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 4

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 4 fuel injector faulty

P0205

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 5

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 5 fuel injector faulty

P0206

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 6

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 6 fuel injector faulty

Diagnostic information - DME (DFI) control unit

Accelerator pedal potentiometer 2

Diagnostic conditions

- Ignition on > 3 second(s)
- then accelerator pedal pressed down fully > 3 second(s)
- then release accelerator pedal fully
- then idling > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in wiring to potentiometer 2
- ♦ Accelerator pedal potentiometer 2 faulty

Fault effects

- ♦ The accelerator pedal angle is calculated from the remaining potentiometer signal.
- ♦ No acceleration initially + increased idle speed (approx. 800 rpm)
 - ▶ until the accelerator pedal is released fully.
- ♦ Then acceleration, with
 - ♦ significantly reduced engine torque (max. approx. 100 ... 180 Nm)
 - ♦ and limited engine speed (max. approx. 5,000 rpm)
- ♦ If the accelerator and brake are pressed at the same time, the engine will run at idle speed.

Diagnostic information - DME (DFI) control unit

Accelerator pedal potentiometer 2

Diagnostic conditions

- Ignition on > 3 second(s)
- then accelerator pedal pressed down fully > 3 second(s)
- then release accelerator pedal fully
- then idling > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in wiring to potentiometer 2
- ♦ Accelerator pedal potentiometer 2 faulty

Fault effects

- ♦ The accelerator pedal angle is calculated from the remaining potentiometer signal.
- ♦ No acceleration initially + increased idle speed (approx. 800 rpm)
 - ▶ until the accelerator pedal is released fully.
- ♦ Then acceleration, with
 - ♦ significantly reduced engine torque (max. approx. 100 ... 180 Nm)
 - ♦ and limited engine speed (max. approx. 5,000 rpm)
- ♦ If the accelerator and brake are pressed at the same time, the engine will run at idle speed.

Diagnostic information - DME control unit

Boost pressure control limit exceeded

INFORMATION

The boost pressure is detected as being too high and is reduced by the boost pressure control system.

The fault is detected if the boost pressure control system cannot adjust the boost pressure correctly within the control limits.

Diagnostic conditions

- Full-load acceleration

Possible fault causes

- ♦ Diverter valve does not open or does not open sufficiently (► see )
- ♦ Insufficient vacuum supply for diverter valve/boost pressure adjuster
- ♦ Boost pressure adjuster does not adjust or does not adjust sufficiently
- ♦ Wrong value from boost pressure sensor to DME control unit

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Engine torque is reduced

INFORMATION

When the diverter valves are in normal position, they are closed by a spring, but can be pressed on by boost pressure.

They each have two connections:

- ♦ a boost pressure connection for closing the valve and keeping it closed (branch ahead of throttle valve);
- ♦ a vacuum connection for opening the valve; switched by a solenoid valve.

P0235

Diagnostic information - DME (DFI) control unit

Boost pressure sensor (pressure sensor ahead of throttle valve) faulty

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Pressure sensor ahead of throttle valve faulty

Fault effects

- ♦ Substitute value from model

P0236

Diagnostic information - DME (DFI) control unit

Boost pressure sensor (pressure sensor ahead of throttle valve) implausible

Diagnostic conditions

- Ignition on > 10 second(s)
- Then: Acceleration with wide-open throttle for as long as possible, immediately followed by an overrun phase lasting as long as possible
- Then engine idling
- None of the following faults stored: P1638, P1639, P0107, P0108, P2228, P2229, P1183, P1184, P1187, P1188, P1189, P1190

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Ground supply faulty
- ♦ Boost pressure sensor faulty

Fault effects

- ♦ Substitute value from model

P0237

Diagnostic information - DME (DFI) control unit

Boost pressure sensor (pressure sensor ahead of throttle valve): electrical fault

Diagnostic conditions

- Ignition on or engine running

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Pressure sensor ahead of throttle valve faulty

Fault effects

- ♦ Substitute value from model

P0238

Diagnostic information - DME (DFI) control unit

Boost pressure sensor (pressure sensor ahead of throttle valve): electrical fault

Diagnostic conditions

- Ignition on or engine running

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Pressure sensor ahead of throttle valve faulty

Fault effects

- ♦ Substitute value from model

P0243

Diagnostic information - DME (DFI) control unit

Boost pressure control cycle valve 1 output stage

Diagnostic conditions

- Engine running
- Boost pressure control cycle valve activated

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit or open circuit in the lines (power supply or boost pressure control cycle valve activation)
- ♦ Boost pressure control cycle valve faulty

P0245

Diagnostic information - DME (DFI) control unit

Boost pressure control cycle valve 1 output stage

Diagnostic conditions

- Engine running
- Boost pressure control cycle valve activated

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit or open circuit in the lines (power supply or boost pressure control cycle valve activation)
- ♦ Boost pressure control cycle valve faulty

P0246

Diagnostic information - DME (DFI) control unit

Boost pressure control cycle valve 1 output stage

Diagnostic conditions

- Engine running
- Boost pressure control cycle valve activated

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit or open circuit in the lines (power supply or boost pressure control cycle valve activation)
- ♦ Boost pressure control cycle valve faulty

P0247

Diagnostic information - DME (DFI) control unit

Boost pressure control cycle valve output stage

Diagnostic conditions

- Engine running
- Boost pressure control cycle valve activated

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit or open circuit in the lines (power supply or boost pressure control cycle valve activation)
- ♦ Boost pressure control cycle valve faulty

P0249

Diagnostic information - DME (DFI) control unit

Boost pressure control cycle valve output stage

Diagnostic conditions

- Engine running
- Boost pressure control cycle valve activated

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit or open circuit in the lines (power supply or boost pressure control cycle valve activation)
- ♦ Boost pressure control cycle valve faulty

P0250

Diagnostic information - DME (DFI) control unit

Boost pressure control cycle valve output stage

Diagnostic conditions

- Engine running
- Boost pressure control cycle valve activated

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit or open circuit in the lines (power supply or boost pressure control cycle valve activation)
- ♦ Boost pressure control cycle valve faulty

P025A

Diagnostic information - DME (DFI) control unit

Fuel pump control unit: activation

- Only for rear-wheel drive vehicles

Diagnostic conditions

- Engine start (electric fuel pump activated)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Open circuit in control line for DME (DFI) control unit ⇒ Fuel pump control unit
- ♦ Faulty power supply or ground supply for fuel pump control unit
- ♦ Fuel pump control unit faulty

• After work carried out previously:

- ♦ Connectors interchanged (fuel pump control unit ◀▶ electric fuel pump)
- i** The connector with 2 cables must be connected to the fuel pump flange, while the connector with 5 cables must be connected to the fuel pump control unit!

P025A

Diagnostic information - DME (DFI) control unit

Fuel pump control unit: activation

- Only for all-wheel drive vehicles

Diagnostic conditions

- Engine start (electric fuel pump activated)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Open circuit in control line for DME (DFI) control unit ⇒ Fuel pump control unit
- ♦ Faulty power supply or ground supply for fuel pump control unit
- ♦ Fuel pump control unit faulty

P025C

Diagnostic information - DME (DFI) control unit

Fuel pump control unit: activation

- Only for rear-wheel drive vehicles

Diagnostic conditions

- Engine start (electric fuel pump activated)

Possible fault causes

- ♦ Fault in electric fuel pump area (See ⓘ)
- ♦ Short circuit to ground in control line for DME (DFI) control unit ⇒ Fuel pump control unit
- ♦ Fuel pump control unit faulty

ⓘ INFORMATION

- ♦ If the fuel pump control unit detects an external or internal fault, it connects the control line to ground for a certain time.
- ♦ This short circuit to ground is recognized as the fault type >>Below limit value<<, just like an actual short circuit to ground in the control line.

P025C

Diagnostic information - DME (DFI) control unit

Fuel pump control unit: activation

- Only for all-wheel drive vehicles

Diagnostic conditions

- Engine start (electric fuel pump activated)

Possible fault causes

- ♦ Short circuit to B+ in control line for DME (DFI) control unit ⇒ Fuel pump control unit
- ♦ Short circuit to ground in control line for DME (DFI) control unit ⇒ Fuel pump control unit
- ♦ Fuel pump control unit faulty

P025D

Diagnostic information - DME (DFI) control unit

Fuel pump control unit: activation

- Only for rear-wheel drive vehicles

Diagnostic conditions

- Engine start (electric fuel pump activated)

Possible fault causes

- ♦ Short circuit to B+ in control line for DME (DFI) control unit ⇒ Fuel pump control unit
- ♦ Fuel pump control unit faulty

P025D

Diagnostic information - DME (DFI) control unit

Fuel pump control unit: activation

- Only for all-wheel drive vehicles

Diagnostic conditions

- Engine start (electric fuel pump activated)

Possible fault causes

- ♦ Short circuit to B+ in control line for DME (DFI) control unit ⇒ Fuel pump control unit
- ♦ Short circuit to ground in control line for DME (DFI) control unit ⇒ Fuel pump control unit
- ♦ Fuel pump control unit faulty

P0261

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 1

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 1 fuel injector faulty

P0262

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 1

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 1 fuel injector faulty

P0264

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 2

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 2 fuel injector faulty

P0265

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 2

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 2 fuel injector faulty

P0267

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 3

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 3 fuel injector faulty

P0268

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 3

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 3 fuel injector faulty

P0270

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 4

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 4 fuel injector faulty

P0271

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 4

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 4 fuel injector faulty

P0273

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 5

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 5 fuel injector faulty

P0274

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 5

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 5 fuel injector faulty